# Regime-Adaptive Supervisory Governance for Instability-Aware Portfolio Stabilization

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**Abstract.** Portfolio optimization performs best when the risk structure used to construct a portfolio is reasonably stable. However, in stressed market regimes, covariance relationships, volatility levels, and cross-asset correlations can move in tandem, making conventional allocation rules less robust. In this paper, we propose an interpretable regime-adaptive supervisory governance framework for portfolio stabilization during market instability. The proposed framework integrates covariance drift, rolling volatility, and correlation stress into a general instability index and flexibly adjusts governance through concentration limits and regularization strength. Instead of switching allocation models, the approach uses a unified optimization framework to adapt governance behavior as instability increases. We empirically test the framework on 18 liquid U.S. stocks from 2005 to 2025 and find that it maintains broad diversification, slightly improves selected downside-risk measures relative to equal weight, and remains transparent in out-of-sample testing while accounting for transaction costs. Ablation studies show that governance regularization and concentration control are the primary stabilizing factors, whereas regime switching only marginally improves performance in the tested setting. We find no evidence of alpha generation or predictive outperformance. The contribution is therefore more limited and feasible: instability-aware supervision as a governance layer for robust portfolio control.

**Keywords:** Portfolio optimization · Covariance instability · Supervisory governance · Regime adaptation · Robust diversification · Multi-agent system (MAS)-compatible architecture

## 1 Introduction

Portfolio optimization concerns not only how to choose weights but also how much freedom the optimizer should have as the market environment changes. In tranquil markets, historical estimates of risk and dependence may be sufficiently stable for practical allocation. The same estimates can become fragile in stressed markets. Covariance matrices shift, volatility increases, diversification benefits decline, and portfolios that appeared diversified under normal conditions can become more concentrated in their effective risk exposure.

The classic version of mean-variance optimization utilizes the framework developed by Markowitz [22] and remains a key part of portfolio construction. The problem with the classic version, however, is well known, namely, that it is very sensitive to noisy input data. Therefore, robust estimation of the covariance matrix, shrinkage, and diversified portfolio allocations are all relevant considerations for real-world risk management [18, 23, 8]. Another important takeaway from studies into regime-sensitive asset allocation is that markets are dynamic and portfolio policies must adapt accordingly [3, 12, 30].

But recently conducted researches in artificial intelligence in portfolio management tend to focus either on return prediction, reinforcement learning, or black-box optimization. These research lines are important, but sometimes not easy to implement because of the need for interpretability, stability, and auditability. In this paper, we take another approach. We neither construct the alpha model nor build return predictors or autonomous trading agents. Instead, we ask another question: is it possible to build an interpretable supervisory module that will be able to change portfolio governance during market instability?

This particular framework can be referred to as a governance-driven stabilization tool since it takes into account covariance shifts, rolling volatility, and correlation stress and converts them into an instability index. Based on the regime state, it adjusts concentration limits and regularization strength. The optimizer has more liberty during periods of stability, while instability requires the framework to become more conservative.

The main contributions are:

1. A regime-adaptive governance regularization mechanism that modifies concentration limits and regularization strength, rather than replacing the underlying optimization model.
2. An instability-aware supervisory signal that integrates covariance drift, rolling volatility, and correlation stress into a single interpretable governance trigger.
3. An empirical evaluation with expanding normalization, out-of-sample testing, transaction costs, sensitivity analysis, ablation models, and concentration diagnostics with a robustness focus.

The framework is intended to be modular, but the implementation is centralized. Its monitoring, governance adaptation, optimization, and audit components can be naturally mapped to a Multi-Agent supervisory architecture in the general sense of role-based system decomposition and coordinated decision modules [32]. This is a claim of architectural compatibility, not a claim that the present study realizes distributed autonomous agents.

## 2 Related Work

Mean-variance optimization [22] and equilibrium risk pricing [29] form the basis of modern portfolio theory. These frameworks provide insight into the relationship between risk and expected return, but their inputs are difficult to estimate accurately. Shrinkage covariance estimation addresses part of this issue by improving the conditioning of high-dimensional covariance matrices [18]. The practical significance of estimation limitations and robustness is discussed in more detail in the context of risk allocation [23].

Another central concern is the difficulty of persistently beating simple diversification. DeMiguel et al. [8] show that naive 1/N diversification is hard to beat out of sample. A diversification-oriented approach to portfolio construction also implies an emphasis on concentration and out-of-sample robustness [27]. Such results point to the need for a cautious attitude towards the assessment process: a governance mechanism is to be assessed based not on the direct increase in the portfolio performance, but on its ability to ensure diversification and to manage concentration and downside risk behavior.

Regime-dependent portfolio management has a rich tradition. Financial connections are subject to changes through time [19]. Regime-switching models have been used for international allocation, dynamic correlations, and multivariate asset allocation [3, 25, 12]. Hidden Markov and switching-control frameworks further demonstrate how systems can adapt under changing latent states [9, 24]. Work on volatility regimes [13, 10] provides theoretical grounding for detecting market stress conditions, while adaptive responses to market classifications [15] emphasize changing allocation behavior. Furthermore, recent work combines regime switching with robust portfolio optimization, random matrix theory [17], GARCH dynamics, and asymmetric dependence [14, 26, 30]. The deterioration of diversification benefits under stress [20] motivates the correlation-stress channel in our instability index.

Robust portfolio optimization also informs the proposed method. Robust covariance estimators and minimum-variance formulations address instability in estimated risk structure [33, 7]. Conditional value-at-risk provides a coherent way to examine downside loss behavior [28], while recent robust formulations incorporate covariance uncertainty, CVaR constraints, and active-management settings [34, 16].

The present paper differs from return-predictive AI portfolio systems. Deep reinforcement learning studies provide useful context for adaptive portfolio research [11], but the proposed framework is intentionally more modest. It is an interpretable supervisory governance layer designed to stabilize allocation behavior under covariance instability.

# 3 Methodology

## 3.1 Asset Universe and Returns

In the empirical universe, there are 18 liquid U.S. stocks in technology, finance, healthcare, consumer goods, industrials, energy, internet, retail, semiconductors, and utility sectors. Daily adjusted close prices are recorded from 1 January 2005 to 1 January 2025. The daily log returns are calculated as

$$r_{i,t} = \log\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

where $P_{i,t}$ is the adjusted price of asset $i$ at time $t$. Equation (1) therefore defines the daily return input used throughout the monitoring and optimization stages.

## 3.2 Instability Monitoring

The supervisory monitor is constructed from three instability channels. The first channel is covariance drift which captures changes in the rolling covariance matrix as follows:

$$D_t = \|\Sigma_t - \Sigma_{t-1}\|_F, \tag{2}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\Sigma_t$ is the rolling covariance matrix. Thus, Equation (2) measures the period-to-period change in estimated covariance structure. According to random matrix theory [21], covariance matrices contain some noise, which makes noise-aware covariance estimation important. The second channel is rolling volatility, which measures the volatility pressure on the cross-assets and idiosyncratic risk shifts [6]. The third channel is correlation stress, which is captured by the average pairwise correlation level. It shows the tendency of disappearing diversification effects under stress [20].

The process of normalizing the signals involves the use of an expanding mean and standard deviation, but with a lag of one period to prevent any data leakage:

$$Z_{k,t} = \frac{X_{k,t} - \mu^{\exp}_{k,t-1}}{\sigma^{\exp}_{k,t-1}}, \tag{3}$$

where $X_{k,t}$ is the raw value of channel $k$. Equation (3) gives each channel a comparable, lagged standardized scale. The composite instability index is then

$$I_t = \alpha Z_{D,t} + \beta Z_{V,t} + \gamma Z_{C,t}, \tag{4}$$

with $\alpha = 0.4$, $\beta = 0.3$, and $\gamma = 0.3$. Equation (4) aggregates the three monitored channels into the supervisory signal. A regime is classified as unstable when $I_t > \tau$, with $\tau = 1.0$ in the main experiment.

### 3.3 Governance-Adaptive Optimization

The portfolio is optimized using a Ledoit-Wolf covariance estimate [18], informed by random matrix theory applications [5]. The framework solves for each rebalance date

$$\min_{w} \sqrt{w^\top \Sigma w} + \lambda_g \sum_{i=1}^{N} w_i^2, \tag{5}$$

subject to

$$\sum_{i=1}^{N} w_i = 1, \qquad 0 \le w_i \le u_g. \tag{6}$$

where $w$ is the vector of portfolio weights, $\Sigma$ is the estimated covariance matrix, $\lambda_g$ is the strength of governance regularization, and $u_g$ is the maximum asset weight in regime $g$. Equation (5) trades portfolio risk against concentration, while Equation (6) imposes full investment, long-only weights, and the regime-specific concentration cap.

The stable regime uses $u_g = 0.35$ and $\lambda_g = 1$. The unstable regime uses $u_g = 0.20$ and $\lambda_g = 10$. Regularization methods [31] and concentration limits [4] are well-established for stabilizing optimization under uncertainty. The design therefore avoids sudden switching between unrelated models. It keeps the optimizer structurally consistent, but changes the concentration risk that the governance layer allows. This leads to a supervisory pipeline covering the market data, instability monitoring, governance adaptation, Ledoit-Wolf constrained optimization, and audit storage.

### 3.4 Backtesting Protocol

The backtest uses a 252-day training window and rebalances every 21 days. The out-of-sample performance of portfolios is done from 1 January 2015 to 1 January 2025. The transaction cost is computed based on proportional transaction costs at a level of 0.001. This reflects typical transaction cost considerations [2]. The measures reported include annual returns, annual volatility, Sharpe ratio, maximum drawdown, conditional value at risk at 95%, Herfindahl Hirschman Index (HHI), turnover, threshold sensitivity, and ablation analysis.

HHI is used for concentration analysis:

$$\mathrm{HHI}_t = \sum_{i=1}^{N} w_{i,t}^2. \tag{7}$$

Lower HHI indicates broader diversification, and Equation (7) is used to calculate the concentration results reported below. The effective number of assets is 1/HHI.

Table 1: Continuous crisis-window instability test.

| Sample | $N$ | Mean $I_t$ | $I_t > \tau$ |
|---|---|---|---|
| Crisis windows | 131 | 1.3521 | 38.2% |
| Non-crisis periods | 4585 | -0.0200 | 9.9% |

Welch test: $t = 7.47$, $p = 1.01 \times 10^{-11}$.

Table 2: Out-of-sample performance, 2015–2025.

| Metric | Governance | Equal weight |
|---|---|---|
| Annual return | 0.1665 | 0.1709 |
| Volatility | 0.1815 | 0.1842 |
| Sharpe ratio | 0.8071 | 0.8190 |
| Maximum drawdown | -0.3590 | -0.3601 |
| CVaR (95%) | -0.0279 | -0.0283 |

# 4 Empirical Results

## 4.1 Instability and Regime Activation

The instability index jumps sharply during a number of major stress periods, most notably around the Global Financial Crisis and the COVID-19 crash. In the full signal sample, 4213 observations are classified as stable and 503 as unstable, corresponding to 89.3% stable and 10.7% unstable observations. A direct event check classifies 4 of 9 selected crisis dates as unstable. This binary count is useful but incomplete, because a one-day classification can miss stress that builds over a surrounding window. To test the signal more directly, the mean instability score is compared inside $\pm 10$ calendar-day windows around the selected crisis dates against all remaining observations. Overlapping observations across clustered event windows are de-duplicated before testing. Table 1 shows that crisis windows have substantially higher instability scores than non-crisis periods.

## 4.2 Governance Versus Equal Weight

Table 2 compares the governance framework with the equal-weight benchmark during the out-of-sample period, and Figure 1 illustrates the corresponding regime-annotated equity curve. Equal weight delivers a slightly higher annual return and Sharpe ratio. The governance framework delivers slightly lower volatility, maximum drawdown, and CVaR. Thus, the result does not indicate return dominance. Instead, it indicates stabilization: the framework sustains broad participation and slightly improves some measures of downside risk.

The contrast between stable and unstable regimes is economically meaningful and is supported by regime-level diagnostics. During turbulent periods, the

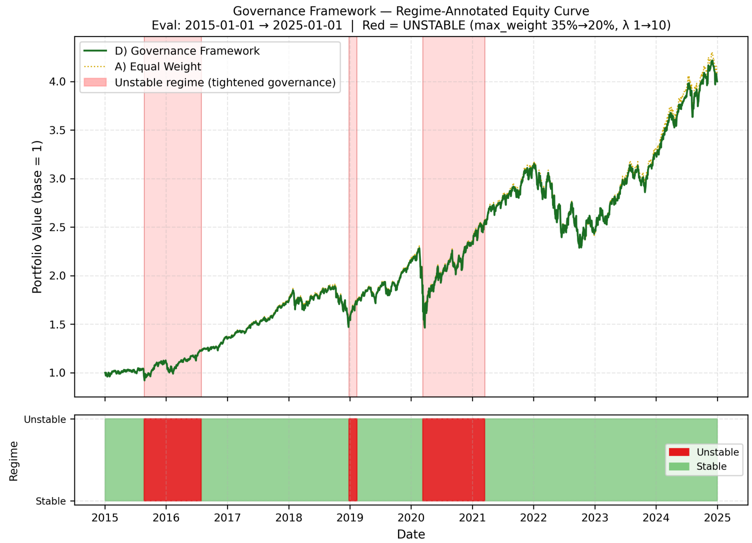



Fig. 1: Regime-annotated equity curve during the evaluation period. Red regions indicate unstable-governance activation.

average daily return declines to 0.000261 from 0.000551. Volatility increases from 0.010 to 0.026, the Sharpe ratio turns negative (-0.094 versus +0.715), and CVaR deteriorates from -0.025 to -0.063.

### 4.3 Concentration and Turnover

The governance framework remains highly diversified. Its mean HHI is 0.0557, close to the equal-weight HHI of 0.0556, indicating an effective number of approximately 18 assets. Table 3 illustrates the importance of this result. The effective number of assets for the minimum-variance baseline is only 5.6, indicating substantially greater concentration, whereas both governance variants remain close to equal-weight diversification. This outcome is a direct consequence of the regularization term $\lambda_g \sum_i w_i^2$, which penalizes concentration and promotes a broadly distributed weight profile while retaining a risk-aware covariance estimate. Thus, the framework combines a risk-aware covariance estimator with a diversified allocation profile. Average turnover per rebalance is 0.0045, and the estimated annual transaction-cost drag

is about 0.0001. These findings are important because they show that the framework does not stabilize the portfolio by making aggressive allocation swings. It remains diversified and low-turnover.

Table 3: Cross-strategy concentration comparison.

| Strategy | Mean HHI | Effective $N$ |
|---|---|---|
| Equal weight | 0.0556 | 18.0 |
| Minimum variance (LW) | 0.1776 | 5.6 |
| Governance, no regime | 0.0557 | 18.0 |
| Governance framework | 0.0557 | 18.0 |

Table 4: Ablation results, out-of-sample period 2015–2025.

| Model | Annual return | Volatility | Sharpe | Max drawdown | CVaR (95%) |
|---|---|---|---|---|---|
| Equal weight | 0.1709 | 0.1842 | 0.8190 | -0.3601 | -0.0283 |
| Minimum variance (LW) | 0.0981 | 0.1463 | 0.5338 | -0.3407 | -0.0218 |
| Governance, no regime | 0.1664 | 0.1812 | 0.8075 | -0.3594 | -0.0279 |
| Governance framework | 0.1665 | 0.1815 | 0.8071 | -0.3590 | -0.0279 |

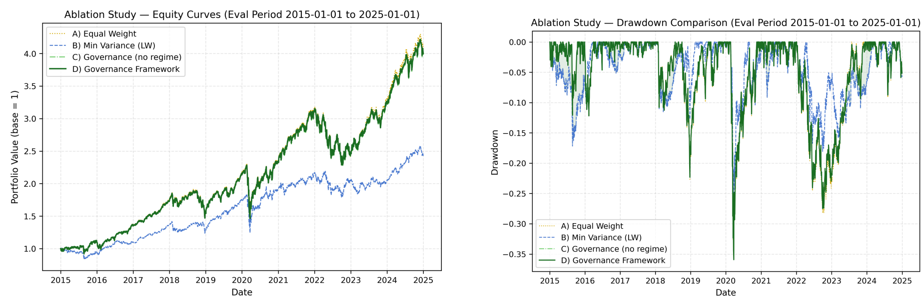
### 4.4 Sensitivity and Ablation

Threshold sensitivity is limited across the tested values of $\tau$: 0.5, 1.0, 1.5, and 2.0. Sharpe ratios remain close to 0.807, while unstable activation falls from 19.8% to 4.0%. This suggests that the regularized governance objective is the main stabilizing component, while the exact threshold has only modest impact within the tested range.

The ablation study compares four models: equal weight, Ledoit-Wolf minimum variance, governance without regime switching, and the full governance framework. Figure 2a presents the corresponding equity paths, Figure 2b presents the drawdown paths, and Figure 2 combines both panels. Table 4 summarises the ablation metrics. Minimum variance reduces volatility and CVaR, but results in substantially lower return and Sharpe. The full framework and the regime switching-free governance are almost indistinguishable. Thus, the evidence is more strongly in favor of governance regularization and concentration control than a large incremental benefit from regime switching.

### 4.5 Statistical Evidence

Paired tests indicate that the governance and equal-weight daily return series differ, with a paired t-test p-value of 0.0168 and a Wilcoxon p-value of 0.0037. This result does not indicate superior performance because equal weight has a slightly higher Sharpe ratio and annual return. Instead, the statistical difference shows that the governance mechanism changes the return distribution rather than improving returns. A more appropriate interpretation is that the governance layer affects allocation behavior and slightly improves some downside-risk measures.




(a) Equity curves.

(b) Drawdown comparison.

Fig. 2: Ablation study: (a) equity curves and (b) drawdown comparison for the four models, evaluation period 2015–2025.

## 5 Discussion

The most notable result is concentration stability: the portfolio remains close to equal-weight HHI while using a risk-aware covariance estimator and transaction-cost-aware rebalancing. The other significant finding is the mild downside stability, which is consistent with modern methods of measuring and mitigating systemic risks [1]. The downside stability is shown in terms of volatility, maximum drawdown, and CVaR. It should be noted that the research does not make any claims of alpha generation or return prediction.

MAS-based interpretation of the problem should also be understood correctly. Although it is centralized, the implementation has four natural components that correspond to supervisory modules–a monitor, a governance adapter, an optimizer, and an audit. Such modularity is useful for constructing multi-agent supervisory systems in the future [32]. However, the current study does not implement distributed autonomous agents, agent communication protocols, or multi-agent reinforcement learning.

## 6 Limitations

The present study has several limitations. First, the empirical universe is limited to 18 U.S. equities. Testing a wider range of asset classes, geographies, and historical periods would provide further validation. Second, the regime threshold is simple and fixed, leaving scope for dynamic adjustment. Third, the crisis-event validation is illustrative and requires future formal benchmarking. Fourth, the full framework remains close to equal weight in some periods because the governance objective strongly preserves diversification. Finally, a promising direction is to deploy the current research pipeline as a fully operational multi-agent system.

## 7 Conclusion

This paper proposed a regime-adaptive supervisory governance framework for instability-aware portfolio stabilization. The framework combines covariance drift, rolling volatility, and correlation stress into an interpretable instability index, then adapts portfolio concentration limits and regularization strength under unstable regimes. Empirical results from 2005 to 2025 show that the framework preserves broad diversification and modestly improves downside stability relative to equal weight, while not improving return or Sharpe performance. The ablation study further shows that governance regularization and concentration control are the dominant stabilizing mechanisms; regime switching modulates this behavior but does not materially improve Sharpe performance in the tested setting.

The contribution is therefore not alpha generation. It is a transparent governance mechanism for stabilizing portfolio behavior under covariance instability. The above position is both practical and methodologically sound in that it provides for adaptive supervisory stabilizing without making unproved claims of predictive intelligence or automated trading.

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